

IBM BoB AI INNOVATION HACKATHON 2026

TrialGuard AI

Clinical Trial Risk Monitor & Protocol Deviation Detector

"From protocol deviations to proactive risk prevention"

TEAM FOURLOOPS

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Deterministic compliance | Site risk intelligence | IBM watsonx.ai + Granite | Human-in-the-loop CAPA

The Problem

TrialGuard AI

Clinical trial monitoring today is manual, reactive and late.



High Data Volume

Patient, visit, medication and lab data at scale.



Slow Manual Review

Checking activity against complex protocols takes time.



Safety & Integrity Risk

Deviations affect patient safety and compliance.



Hard to Spot Early

Recurring deviations and high-risk sites surface late.

TODAY'S REACTIVE CYCLE

Data Collected



Issue Occurs



Discovered Late



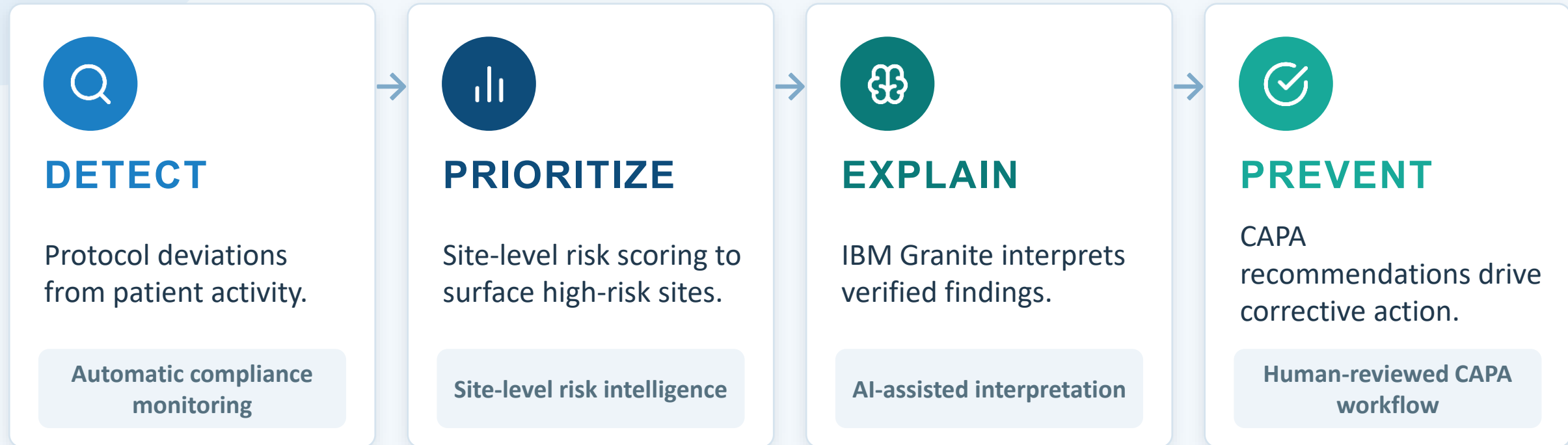
Reactive Fix

TrialGuard AI makes clinical-trial monitoring proactive.

Introducing TrialGuard AI

TrialGuard AI

An AI-powered clinical trial compliance and risk monitoring copilot.



Detect → Prioritize → Explain → Prevent

One continuous loop — from raw trial data to accountable corrective action.

How TrialGuard AI Works

● Rule-Based Compliance

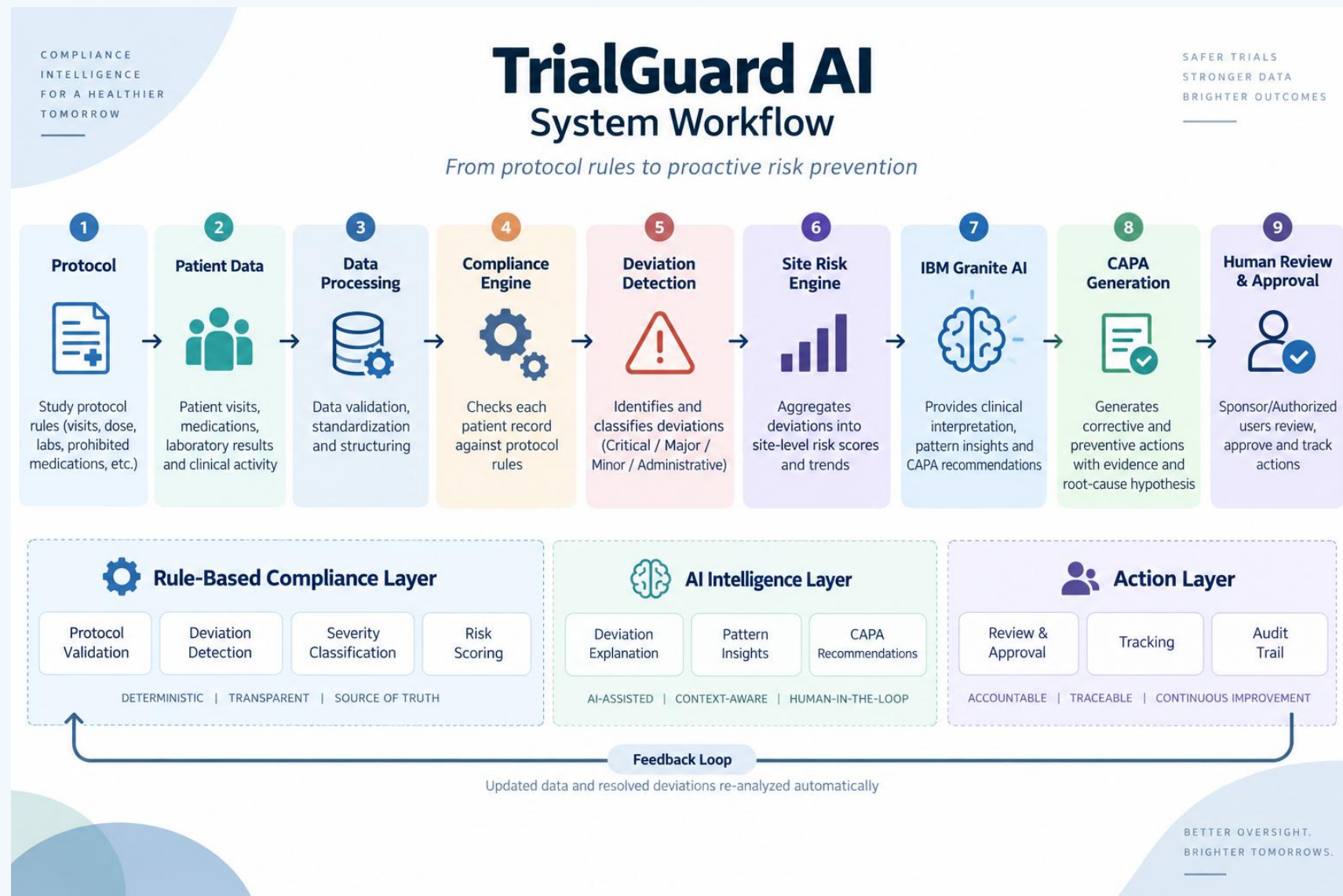
● AI Intelligence

● Action & Governance

Deterministic first.

AI second.

Human always.



Protocol + Patient Data → Compliance → Risk → AI Insight → CAPA → Human Approval

From Protocol Violations to Site Risk

TrialGuard AI



STAGE 1 — DETERMINISTIC COMPLIANCE

- Visit-window violations
- Incorrect dosage
- Prohibited medications
- Missing laboratory assessments
- Missing required data
- Severity classification

Protocol: Visit 2 = Day 14 ± 3

Actual: Day 21

Result: Late Visit → Major Deviation



STAGE 2 — SITE RISK INTELLIGENCE

Patient Deviations



Severity + Frequency + Patterns



Site Risk Score



Low / Medium / High



Priority for Intervention

Deterministic compliance is the source of truth — AI does not invent protocol violations.

AI-Powered Intelligence & Action

TrialGuard AI



IBM GRANITE ON WATSONX.AI

- IBM watsonx.ai provides the AI foundation
- Interprets verified compliance findings
- Explains detected deviations
- Identifies patterns across sites
- Provides AI-assisted recommendations



CAPA WORKFLOW

- Linked to verified deviations
- Evidence and problem context
- Root-cause hypotheses
- Corrective and preventive actions
- Owner, priority, status, approval



Deterministic Engine

Compliance source of truth

IBM Granite

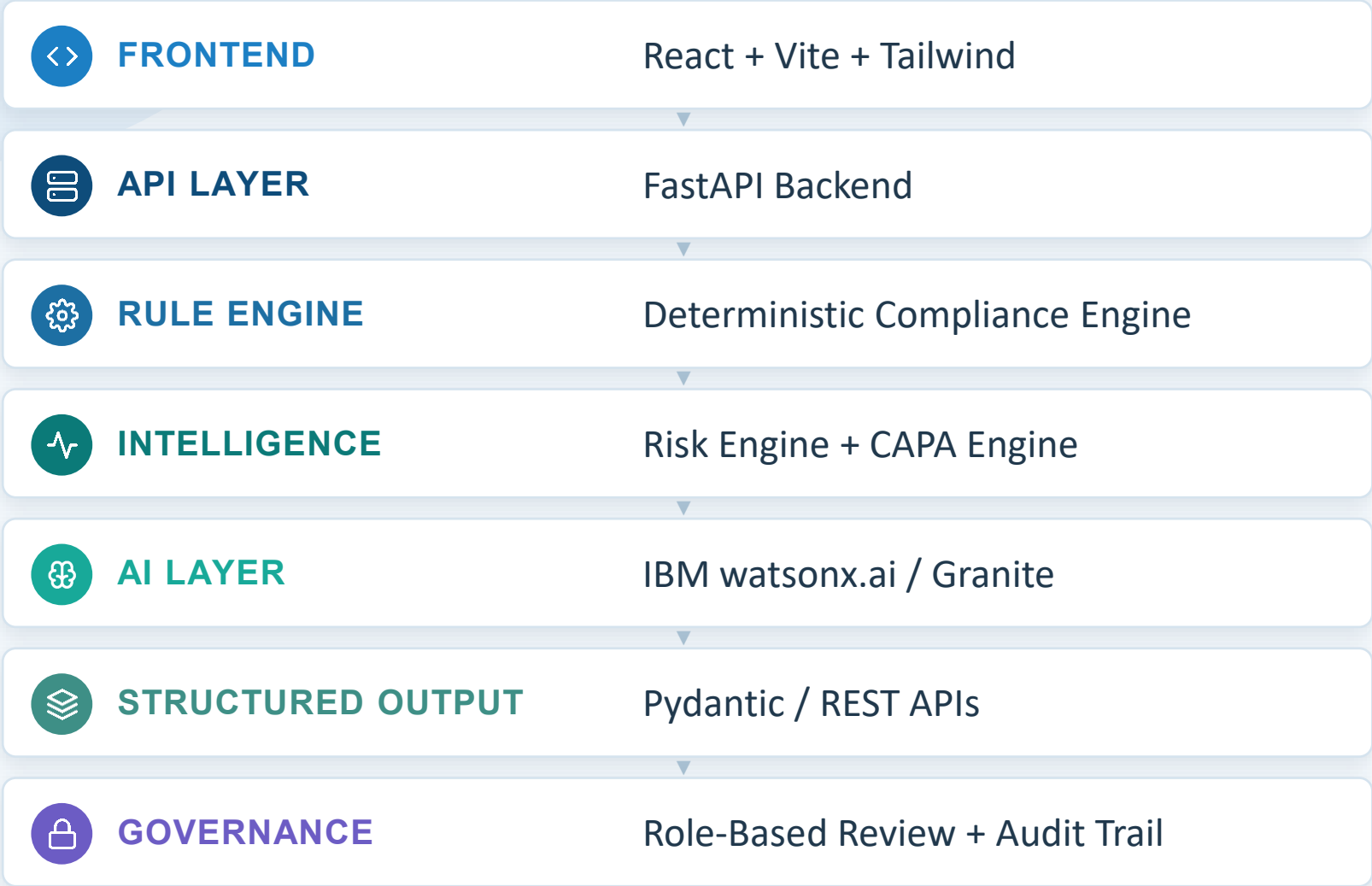
Interpretation & patterns

Human

Review, decision, approval

Technical Architecture

TrialGuard AI



TECHNOLOGY STACK

React

FastAPI

Python

Pydantic

IBM watsonx.ai

IBM Granite

REST APIs



Synthetic clinical-trial dataset

Structured, privacy-safe demo data

Impact & Future Vision

TrialGuard AI



IMPACT

- Faster protocol deviation detection
- Earlier identification of high-risk sites
- Reduced manual monitoring effort
- Consistent compliance assessment
- Faster detection-to-action handoff



FUTURE

- Protocol PDF → automatic rule extraction
- Clinical report → structured patient data
- Continuous incoming-data monitoring
- Advanced predictive risk modelling
- Integration with CTMS platforms

"TrialGuard AI transforms clinical trial monitoring from reactive deviation tracking into proactive risk prevention."

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